

# ONLINE BOOK STORE SQL ANALYSIS

Relational Database Design & Business Analysis Using SQL

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# DATABASE DESIGN

Relational structure for the Online Book Store

Designed a relational Online Book Store database using Books, Customers, and Orders tables.

Tables are connected through primary and foreign keys.

The structure supports sales, customer, product, revenue, and inventory analysis.

## CUSTOMERS

customer\_id (PK)  
customer\_name  
email

## ORDERS

order\_id (PK)  
customer\_id (FK)  
book\_id (FK)  
quantity  
order\_date

## BOOKS

book\_id (PK)  
title  
author  
genre  
price  
stock

# SALES ANALYSIS BY GENRE

## Business Question:

Which book genres generate the highest sales volume?

SQL Concepts: JOIN • SUM() • GROUP BY • ORDER BY

```
SELECT
    b.genre,
    SUM(o.quantity) AS total_books_sold
FROM Books b
JOIN Orders o
    ON b.book_id = o.book_id
GROUP BY b.genre
ORDER BY total_books_sold DESC;
```

## Key Insight:

The result ranks genres by total quantity of books sold, making it possible to identify the highest-volume categories.

# CUSTOMER SPENDING ANALYSIS

## Business Question:

Which customers contribute the most revenue to the bookstore?

SQL Concepts: JOIN • SUM() • GROUP BY • ORDER BY

```
SELECT
    c.customer_name,
    SUM(o.quantity * b.price) AS total_spending
FROM Customers c
JOIN Orders o
    ON c.customer_id = o.customer_id
JOIN Books b
    ON o.book_id = b.book_id
GROUP BY c.customer_id, c.customer_name
ORDER BY total_spending DESC;
```

## Key Insight:

Customer spending analysis identifies high-value customers based on their total purchase value.

# CUSTOMER RANKING USING WINDOW FUNCTIONS

## Business Question:

Who are the highest-spending customers?

SQL Concepts: RANK() • Window Functions • JOIN • GROUP BY • Aggregate Functions

```
SELECT customer_name, total_spending,  
       RANK() OVER (  
         ORDER BY total_spending DESC  
       ) AS customer_rank  
FROM (  
  SELECT c.customer_name,  
         SUM(o.quantity * b.price) AS total_spending  
  FROM Customers c  
  JOIN Orders o ON c.customer_id = o.customer_id  
  JOIN Books b ON o.book_id = b.book_id  
  GROUP BY c.customer_id, c.customer_name  
) AS customer_sales;
```

## Key Insight:

Ranking customers makes it easier to identify the highest-value customers for targeted business strategies.

# BEST-SELLING BOOK BY GENRE

## Business Question:

What is the best-selling book within each genre?

SQL Concepts: CTE • RANK() • PARTITION BY • JOIN • GROUP BY

```
WITH book_sales AS (  
    SELECT b.genre, b.title,  
           SUM(o.quantity) AS total_sold  
    FROM Books b  
    JOIN Orders o ON b.book_id = o.book_id  
    GROUP BY b.genre, b.title  
)  
ranked_books AS (  
    SELECT genre, title, total_sold,  
           RANK() OVER (  
               PARTITION BY genre  
               ORDER BY total_sold DESC  
           ) AS book_rank  
    FROM book_sales  
)  
SELECT genre, title, total_sold  
FROM ranked_books  
WHERE book_rank = 1;
```

## Key Insight:

This analysis identifies the top-performing book within every genre.

# INVENTORY ANALYSIS

Business Question: How much inventory remains for each book after considering ordered quantities?

SQL Concepts: LEFT JOIN • COALESCE() • SUM() • GROUP BY

Business Objective: Calculate ordered quantities and remaining inventory for each book to support stock monitoring.

Why LEFT JOIN? Books with no orders can still be included in the inventory analysis.

Key Insight: Inventory analysis helps identify books with high demand and books that may require restocking.

# REVENUE ANALYSIS BY GENRE

## Business Question:

Which genres contribute the most revenue?

SQL Concepts: JOIN • SUM() • GROUP BY • ORDER BY

```
SELECT
    b.genre,
    SUM(o.quantity * b.price) AS total_revenue
FROM Books b
JOIN Orders o
    ON b.book_id = o.book_id
GROUP BY b.genre
ORDER BY total_revenue DESC;
```

## Key Insight:

Revenue analysis helps identify the book categories that contribute most to overall sales value.



# PROJECT SUMMARY

Business analysis and SQL capabilities demonstrated

| Analysis           | Business Purpose                    |
|--------------------|-------------------------------------|
| Sales by Genre     | Identify high-volume genres         |
| Customer Spending  | Identify high-value customers       |
| Customer Ranking   | Rank customers by spending          |
| Best-Selling Book  | Identify top book within each genre |
| Inventory Analysis | Monitor remaining stock             |
| Revenue by Genre   | Identify high-revenue categories    |

## SQL SKILLS

Basic: SELECT, WHERE, ORDER BY, GROUP BY

Intermediate: INNER JOIN, LEFT JOIN, Aggregate Functions, COALESCE()

Advanced: CTE, RANK(), Window Functions, PARTITION BY

Portfolio focus: sales, customers, products, revenue, inventory